Day 1 – Lab 1.1: Create an Azure Functions App

Module 1 · Serverless & Azure Functions

Objective: Create an Azure Function App using the Flex Consumption hosting plan and prepare the Azure environment for C# development and deployment.

Business scenario: Contoso Orders needs a serverless application to receive and process customer orders without maintaining dedicated application servers.

1. Resource naming

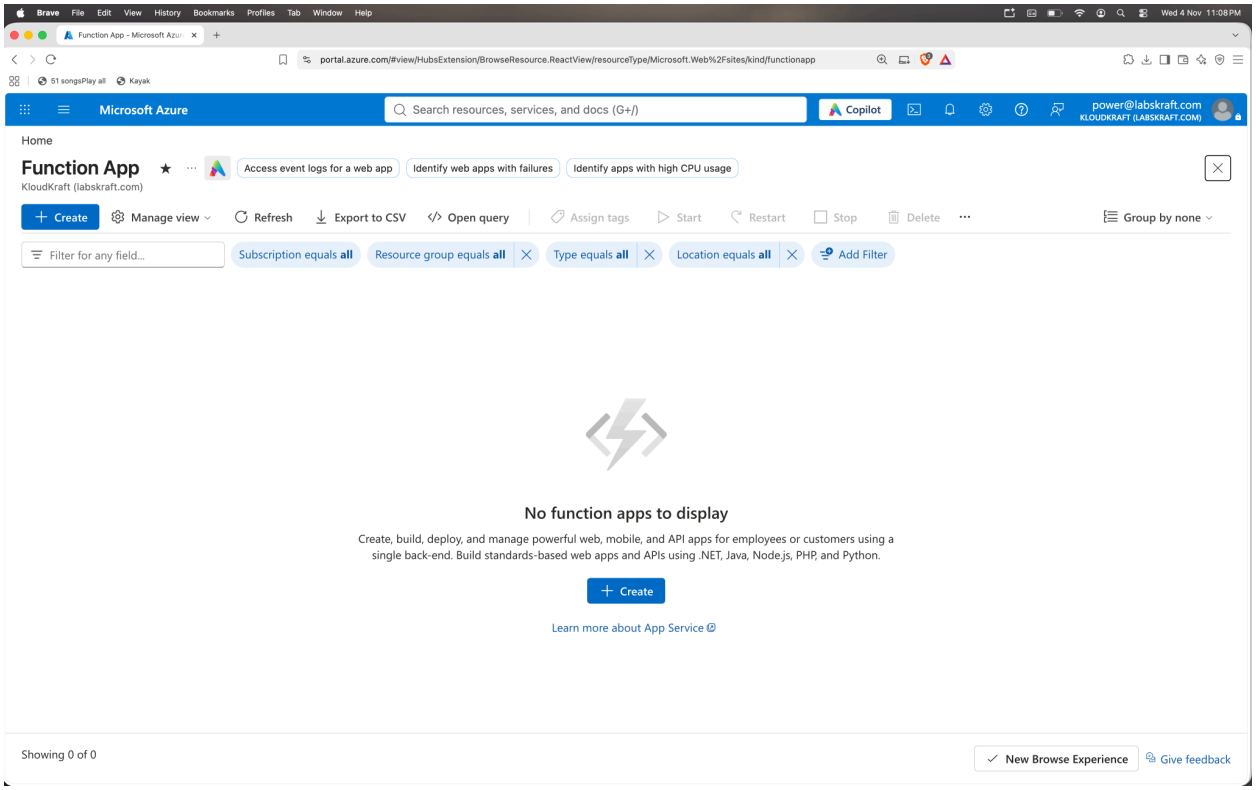
Use the participant ID assigned by the trainer. The example below uses `user01`.

- Function App: `fn-user01-orders`
- Storage Account: `stuser01orders`
- Application Insights: `appi-user01-orders`
- Log Analytics workspace: `law-user01-orders`

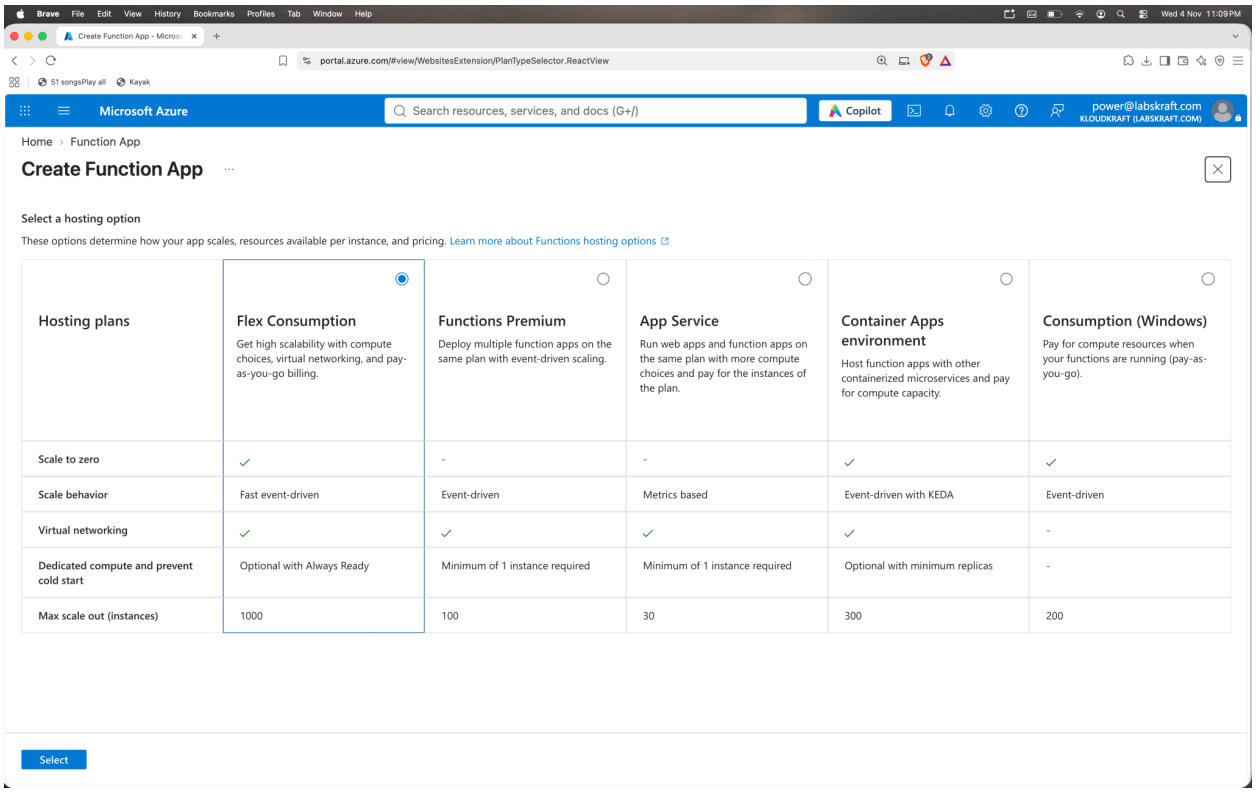
Important: Storage Account names must be globally unique, 3–24 characters long, and contain only lowercase letters and numbers. If the name is unavailable, append a unique numeric suffix.

2. Open the Function App creation wizard

1. Sign in to the Azure Portal.
2. Search for Function App.
3. Select Create.



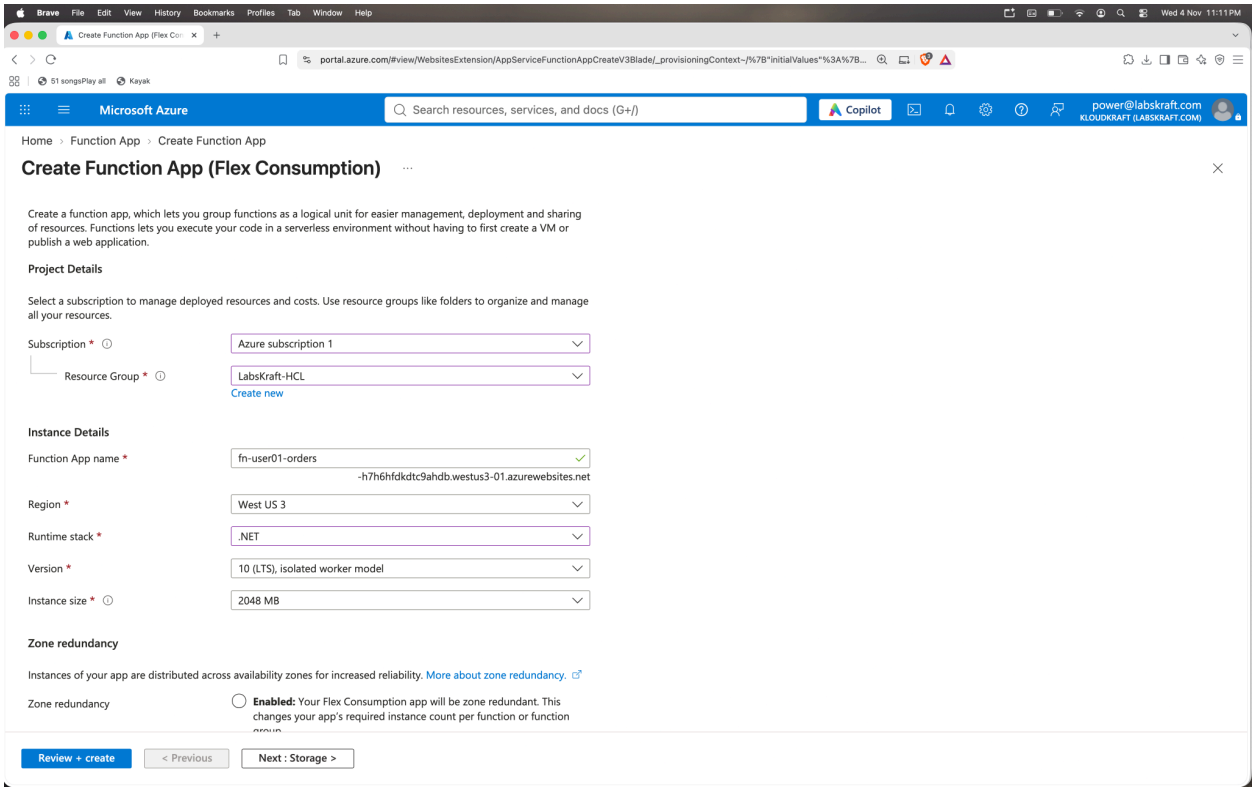
4. Under hosting options, select Flex Consumption.



3. Configure the Function App

On the Basics page, configure the following:

1. Subscription: Select the assigned training subscription.
2. Resource Group: Select the existing shared training Resource Group.
3. Function App name: Enter `fn-user01-orders`.
4. Region: Select the trainer-approved region.
5. Runtime stack: Select `.NET`.
6. Version: Select `.NET 10 (isolated worker)`.
7. Instance size: Keep the default selection.



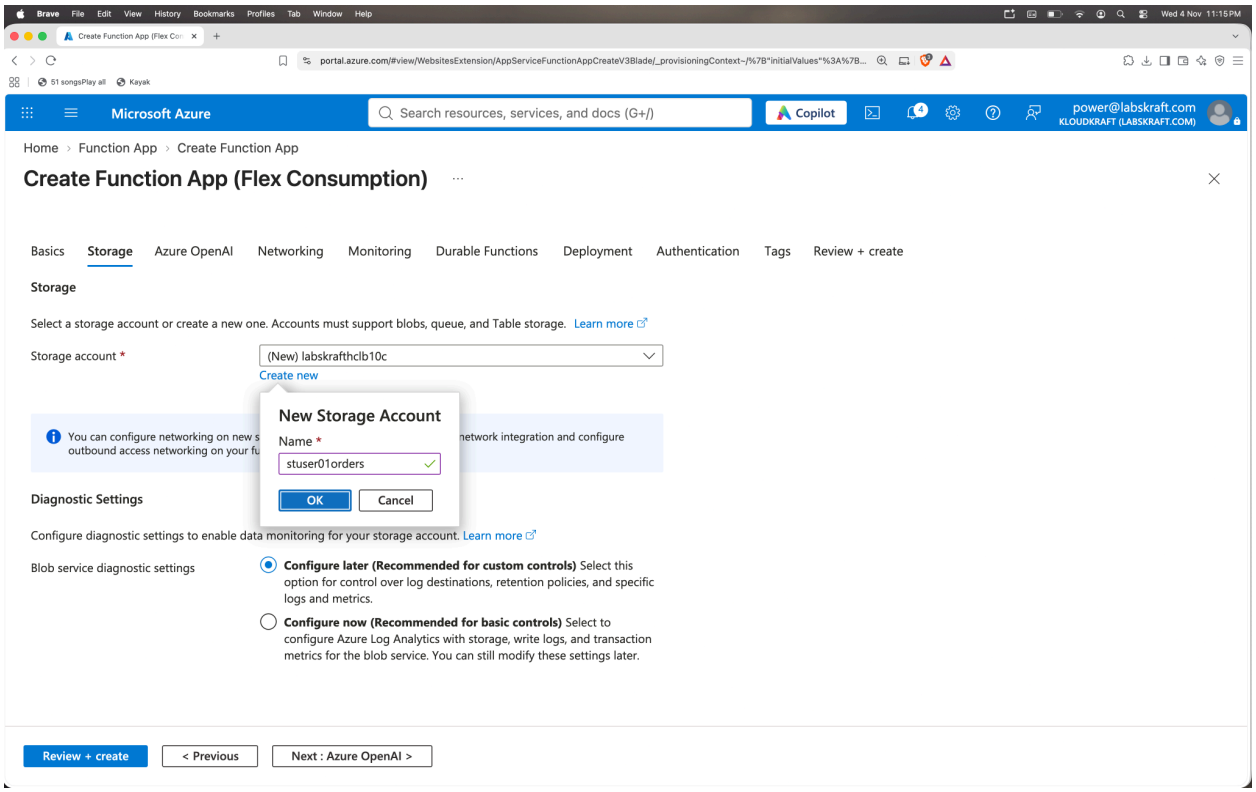
Important: Flex Consumption runs on Linux. C# applications must use the isolated worker model. Do not select the in-process model or legacy Consumption plan.

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4. Configure Storage

1. Open the Storage tab.
2. Select Create new storage account.
3. Enter `stuser01orders`.
4. Select Standard performance.
5. Select Locally-redundant storage (LRS), if presented.
6. Keep the remaining settings at their defaults.

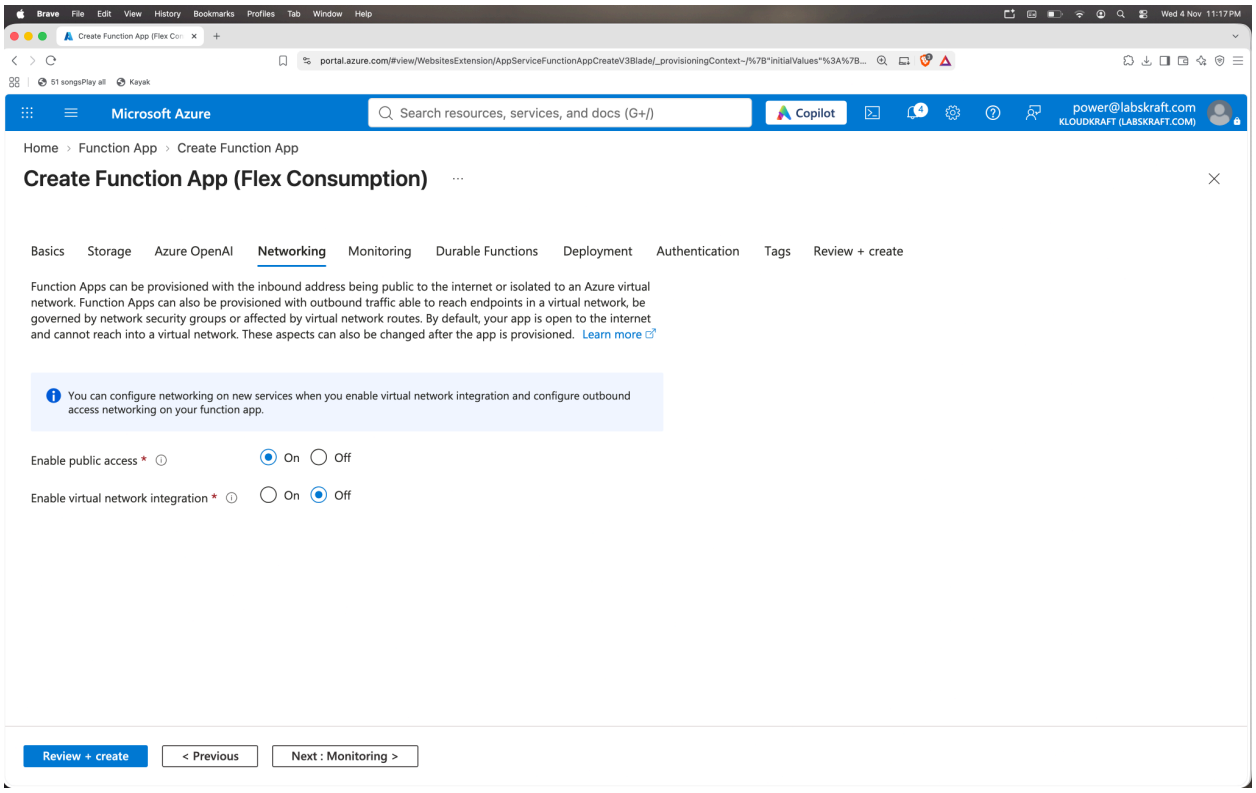
The Storage Account maintains Function runtime state and supports the hosting environment.



Note: If Azure reports that the Storage Account name is unavailable, use a unique suffix, for example `stuser01orders26`.

5. Configure Networking

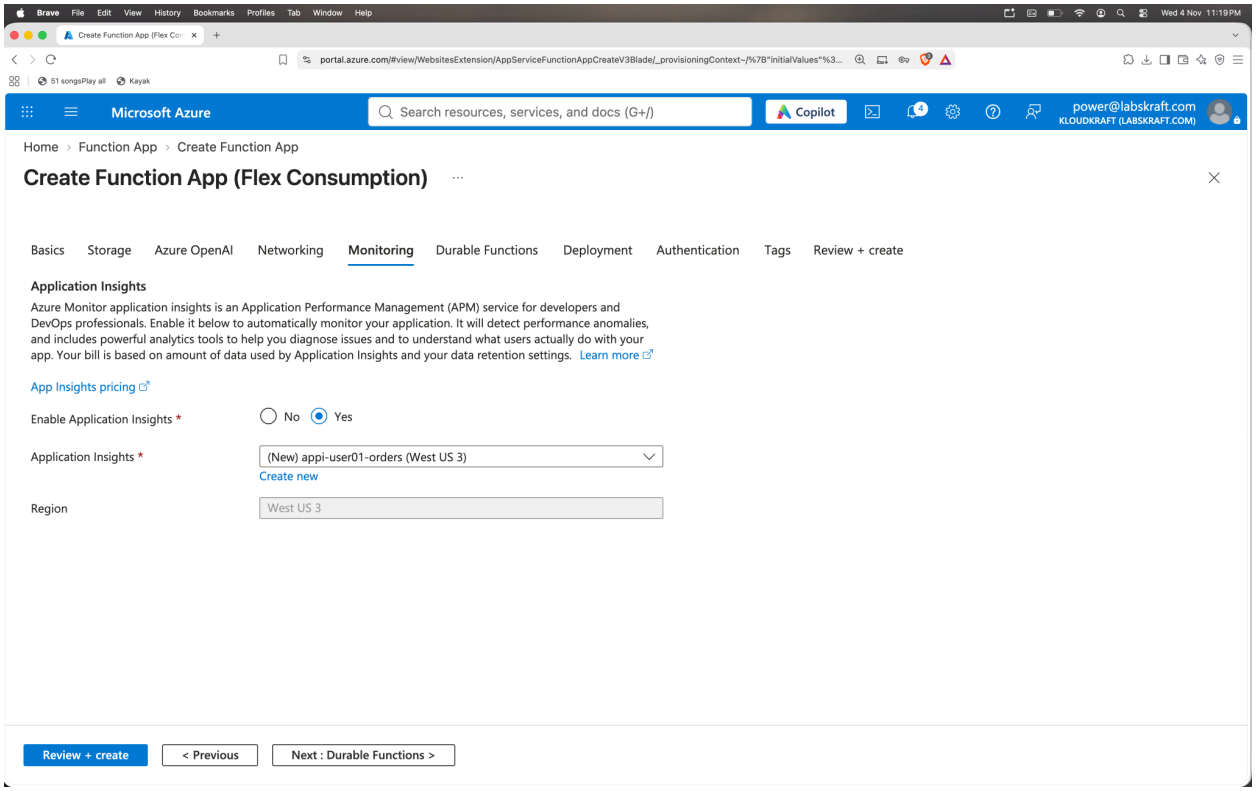
1. Open the Networking tab.
2. Keep public network access enabled for this training exercise.
3. Do not configure VNet integration.
4. Retain the default networking settings.



6. Configure Monitoring

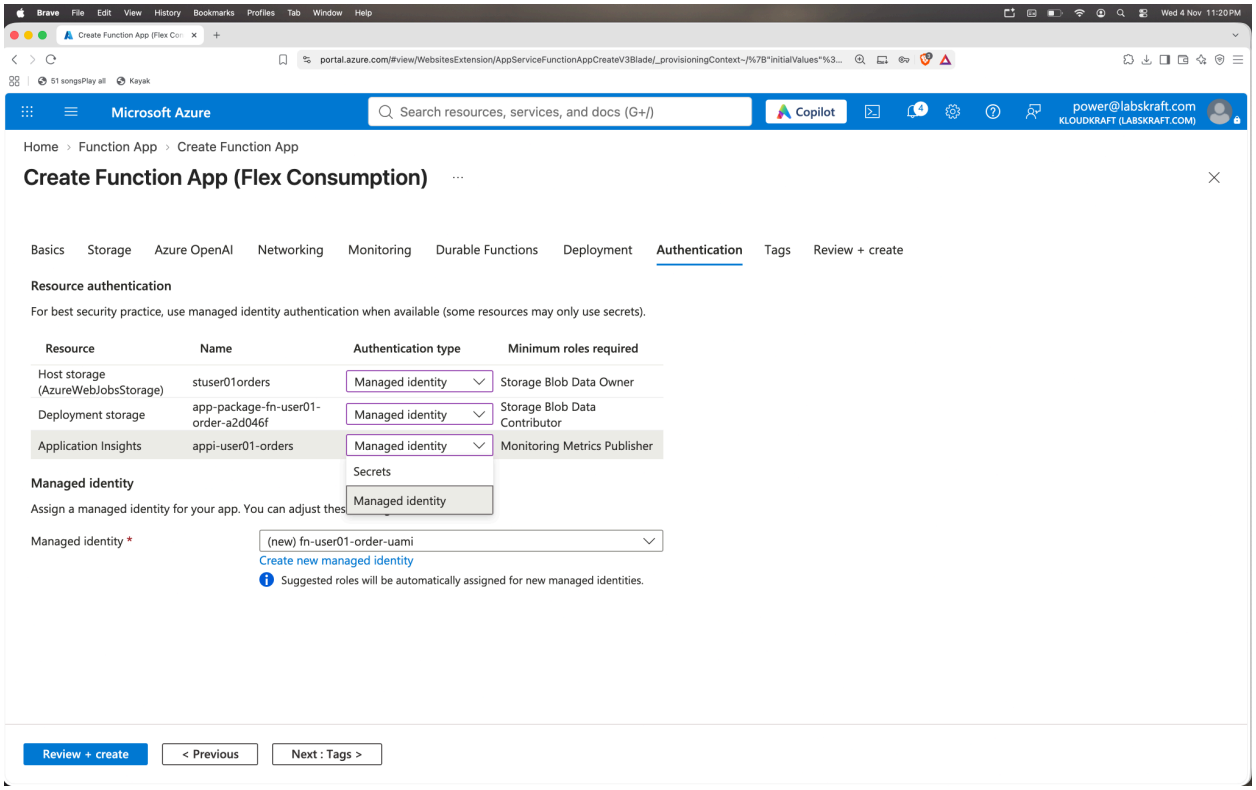
1. Open the Monitoring tab.
2. Ensure Enable Application Insights is selected.
3. Select Create new.
4. Enter `appi-user01-orders`.
5. When prompted, create a new Log Analytics workspace named `law-user01-orders`.
6. Keep the default monitoring configuration.

Application Insights will help us inspect Function executions, requests, traces and exceptions in subsequent labs.



7. Configure Authentication

1. Open the Authentication tab.
2. Select the managed identity authentication option if available in the wizard.



3. Use system-assigned managed identity for this training application.

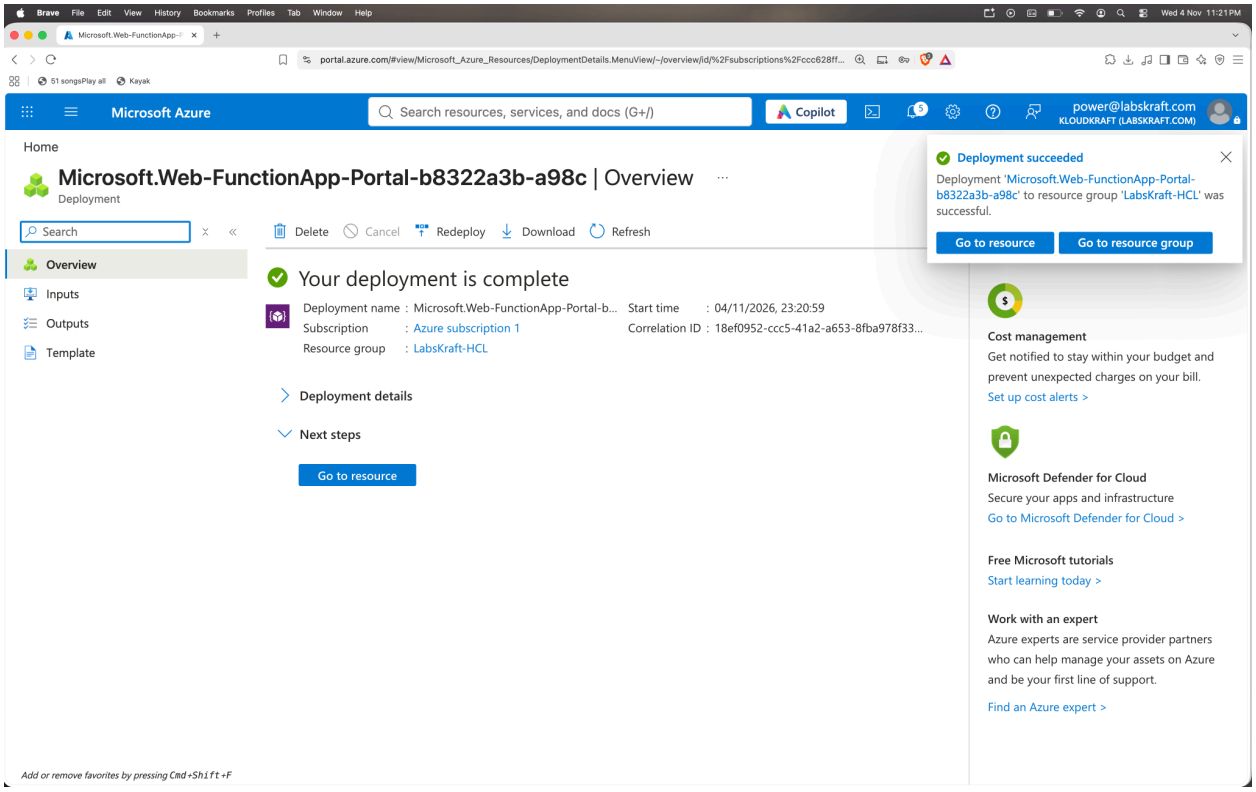
Important: The Function App's managed identity needs appropriate Storage data-plane permissions. If the wizard cannot configure these permissions automatically, the provider/admin must assign the required roles. Do not assume Resource Group Contributor access automatically grants Storage data access.

For the introductory lab, record any permission issue and continue only after the storage authentication configuration is valid.

8. Review and Create

1. Select Review + create.
2. Verify the following:
    - Correct subscription and shared Resource Group.
    - Function App name.
    - Region.

    - Flex Consumption hosting plan.
    - .NET 10 isolated worker runtime.
    - Storage Account.
    - Application Insights.
3. Select Create.
4. Wait for the deployment to complete.
5. Select Go to resource.



Expected result: The Function App is successfully provisioned in the assigned shared Resource Group.

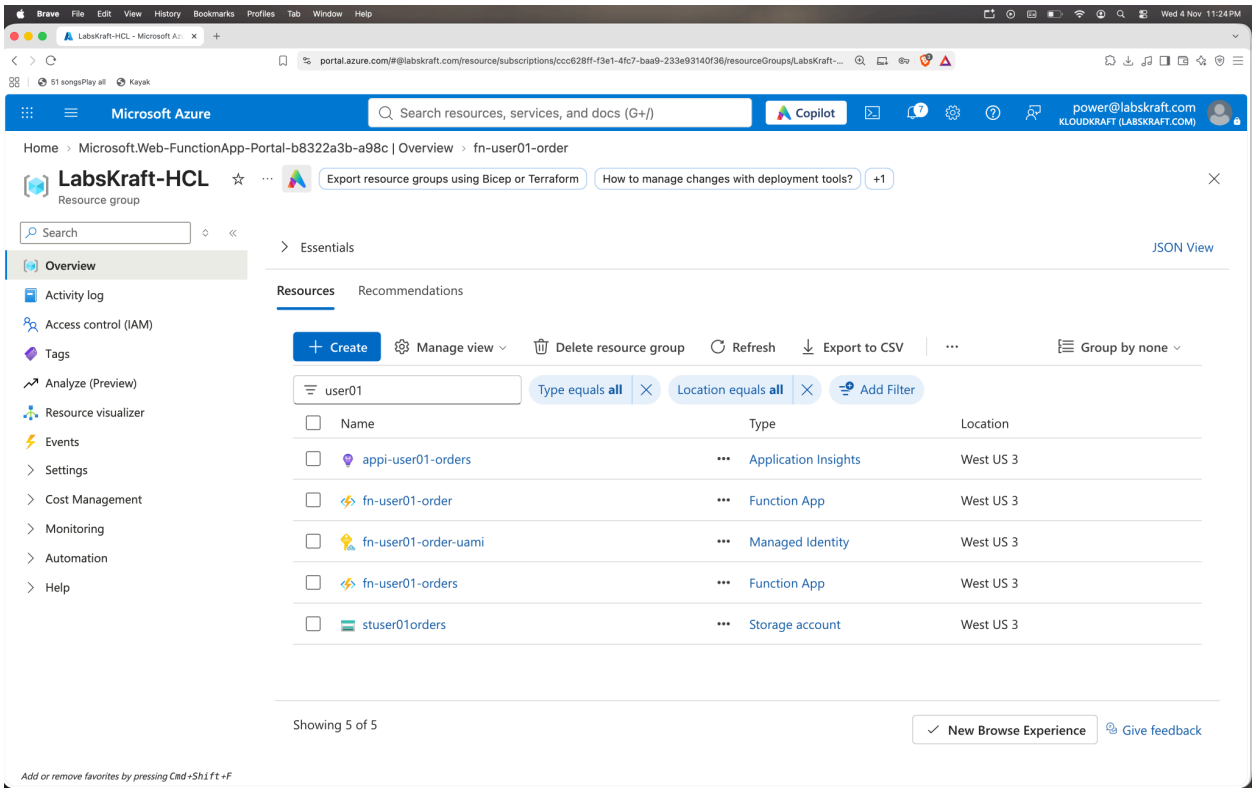
9. Verify the provisioned resources

Navigate to the shared Resource Group.

Confirm that the following resources are available:

- Function App – hosts the C# Functions.

- Flex Consumption plan – provides serverless execution.
- Storage Account – maintains Function runtime storage.
- Application Insights – collects application telemetry.
- Log Analytics workspace – stores and queries monitoring data.



Azure may create additional supporting resources depending on the selected configuration.

10. Verify using Azure CLI

Note: Refer the Azure CLI Guide here

Open the VS Code terminal or PowerShell.

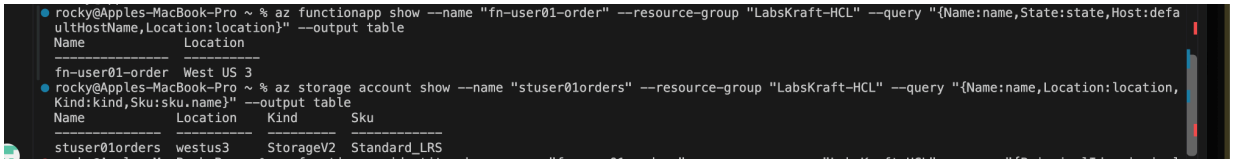
Replace <SHARED_RG> with the actual shared Resource Group name.

Verify Function App

Execute:

```shell
az functionapp show --name "fn-user01-orders"
--resource-group "LabsKraft-HCL" --query
"{Name:name,State:state,Host:defaultHostName,Location
:location}" --output table
```

Expected: Function App name, running state, hostname and region are displayed.



Verify Storage Account

```shell
az storage account show --name "your storage account
name" --resource-group "<SHARED_RG>" --query
"{Name:name,Location:location,Kind:kind,Sku:sku.name}
" --output table
```

Expected: Storage Account details are displayed.

Verify managed identity

If system-assigned identity was enabled:

```shell
az functionapp identity show --name "your functionapp
name" --resource-group "<SHARED_RG>" --query
"{PrincipalId:principalId,TenantId:tenantId}"
--output table
```

Expected: Principal ID and Tenant ID are displayed.



11. Acceptance Criteria

The lab is complete when:

- Function App deployment succeeds.
- Hosting plan is Flex Consumption.
- Runtime is .NET 10 isolated worker.
- Dedicated Storage Account is created.
- Application Insights is enabled.
- Function App hostname is recorded.
- Required storage authentication is configured.
- All resources are visible in the assigned shared Resource Group.

12. Common Issues

Issue	Resolution
Function App name unavailable	Use a unique suffix.
Storage Account name unavailable	Add a numeric suffix.
.NET 10 not listed	Check regional runtime support.

Deployment fails	Check permissions, regional quota and provider registration.
Managed identity cannot access Storage	Verify required data-plane RBAC assignments.
Application Insights creation fails	Check workspace and monitoring resource permissions.

Trainer consideration: All 30 participants will provision Flex Consumption apps in the same subscription and region. Check the regional subscription memory quota before the workshop because Flex Consumption apps share this quota.